1+2)

	Units	25	TV	BV	D	SP	TL1	W
Pressure	bar	3	2.6	1.33	1	1.25	1.08	1.52
Molar Vapor Fraction		0	1	1	0	0	0	0
Molar Liquid Fraction		1	0	0	1	1	1	1
Molar Solid Fraction		0	0	0	0	0	0	0
Mass Vapor Fraction		0	1	1	0	0	0	0
Mass Liquid Fraction		1	0	4.40071e-07	1	1	1	1
Mass Solid Fraction		0	0	0	0	0	0	0
Molar Enthalpy	cal/mol	4688.85	19615	14220	13615.6	6824.53	9032.32	294.815
Mass Enthalpy	cal/gm	49.0841	232.651	152.301	174.149	74.0898	103.864	2.77872
Molar Entropy	cal/mol-K	-75.024	-37.6717	-51.1456	-54.494	-70.1442	-63.8688	-89.2361
Mass Entropy	cal/gm-K	-0.785371	-0.446819	-0.547784	-0.696998	-0.761514	-0.734438	-0.841078
Molar Density	mol/cc	0.0080978	7.80885e-05	4.18402e-05	0.0104207	0.00837076	0.00910199	0.00696845
Mass Density	gm/cc	0.773558	0.00658371	0.00390654	0.814732	0.771043	0.791535	0.739333
Enthalpy Flow	cal/sec	217000	530652	487828	151646	56487.2	101614	2465.57
Average MW		95.5269	84.3108	93.3681	78.1838	92.1115	86.9628	106.097
+ Mole Flows	**kmol/hr**	**166.608**	**97.3924**	**123.5**	**40.0956**	**29.7975**	**40.5**	**30.1072**
+ Mole Fractions								
+ Mass Flows	**kg/hr**	**15915.5**	**8211.23**	**11531**	**3134.83**	**2744.69**	**3521.99**	**3194.29**
− Mass Fractions								
BENZENE		0.00753617	0.517569	0.00663056	0.994107	0.00299651	0.331862	2.19296e-11
TOLUENE		0.71391	0.481883	0.884845	0.00589255	0.995313	0.667737	0.00434227
M-XYLENE		0.278553	0.000548541	0.108525	1.18111e-08	0.0016903	0.000400897	0.995658
Volume Flow	l/min	342.908	20786.7	49195.2	64.1281	59.3286	74.1595	72.0083

Basis Mole

Condenser / Top stage performance

Name	Value	Units
Temperature	79.8168	C
Subcooled temperature		
Heat duty	-2077.68	kW
Subcooled duty		
Distillate rate	40.0956	kmol/hr
Reflux rate	204.967	kmol/hr
Reflux ratio	5.11195	
Free water distillate rate		
Free water reflux ratio		

Reboiler / Bottom stage performance

Name	Value	Units
Temperature	154.741	C
Heat duty	2297.02	kW
Bottoms rate	30.1072	kmol/hr
Boilup rate	232.978	kmol/hr
Boilup ratio	7.73827	
Bottoms to feed ratio		

SIDE STREAM STAGE: 26

VAPOR SIDE STREAM: 123.5 kmol/hr

LIQUID SIDE STREAM: 40.5 kmol/hr

Total number of plates: 53

Streams for the main petlyuk column(2)

Incoming liquid stream:8

Outgoing liquid stream:9

Incoming vapor stream: 35

Outgoing vapor stream: 34

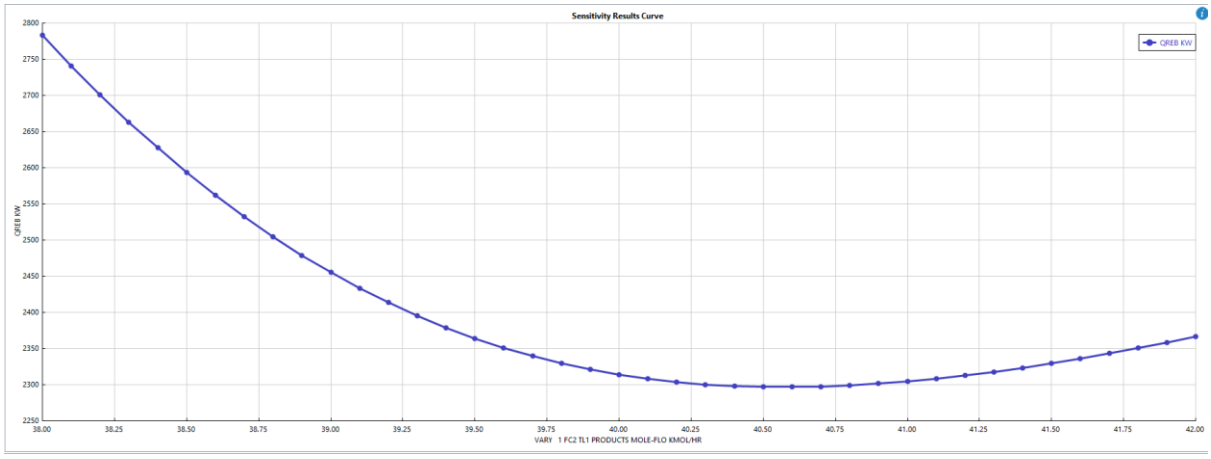
For first column(1)

Feed stage: 20

Distillate rate:40 kmol/hr

Reboiler duty:2800 kW

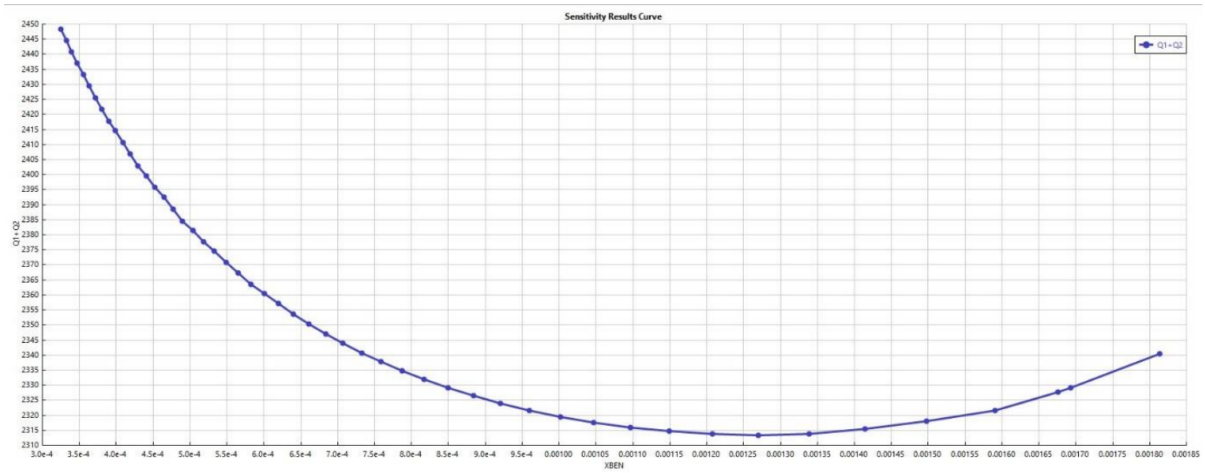
3)From this lab we have reboiler duty of column is:



Column reboiler duty: 2297.02 kW

For previous lab we had direct split more economical heat cost wise.

Column reboiler duty: 2312.5 kW



From energy efficiency perspective petyulk column is better than normal columns(direct split)

NAME: SHRIMI AGRAWAL ROLL NUMBER:221036